

Empirical Software Engineering

Write your answers directly on these pages; there's always a risk that loose papers disappear. Use the back also if possible.

On September 23 at 10.30–11.00 you are welcome to my office (4th floor Jupiter building, Lindholmen), with questions about the grading.

Richard will be at the written exam once at approximately 15.00. He can also be reached later, with permission from the exam invigilator, on 0768-65 33 42.

He who learns must suffer.

—Aeschylus

Grade 3: 33 points; ~50%

Grade 4: 46 points; ~70%

Grade 5: 59 points; ~90%

Maximum: 65 points

Question 1 :

(8p) In *your* opinion, which **steps** are compulsory when conducting Bayesian data analysis? Please **explain** what steps one take when designing models, so that we ultimately can place *some* confidence in the results.

You can either draw a flowchart and explain each step, or write a numbered list explaining each step. (It's ok to write on the backside, if they haven't printed on the backside again...)

Question 2 :

(9p) In Bayesian data analysis we have three principled ways of avoiding overfitting.

The **first** way makes sure that the model doesn't get too excited by the data. The **second** way is to use some type of scoring device to model the prediction task and estimate predictive accuracy. The **third**, and final way is to, if suitable data is at hand, design models that actually do something about overfitting.

Name the three ways. (3p)

Explain and provide examples for each of the three ways (2p+2p+2p)

Question 3 :

(3p) It is possible to view the Normal likelihood as a purely epistemological assumption, rather than an ontological assumption. How would you argue such a statement?

Question 4 :

(8p) **Name** at least four distributions in the exponential family (4p). **Provide examples** of when one can use each distribution when designing statistical models and **explain** what is so special about each distribution you've picked, i.e., their assumptions (4p).

Question 5 :

(5p) Below you see a Generalized Linear Model

$$y_i \sim \text{Binomial}(n, p)$$
$$f(p) = \alpha + \beta x_i$$

What is $f()$ called **and** why is it needed? (2p)

What should $f()$ be in this case? (1p)

Does $f()$ somehow affect your priors? If yes, **provide an example**. (2p)

Question 6 :

(6p) What is the **purpose** and **limitations** of using *Judgement Studies* and *Sample Studies* as a research strategy? Provide **examples**, i.e., methods for each of the two categories, and **clarify** if one use mostly qualitative or quantitative approaches (or both).

Question 7 :

(8p) Below follows an abstract from a research paper. Answer the questions,

- Which of the eight research strategies presented in the ABC framework does this paper likely fit? **Justify and argue!**
- Can you argue the main validity threats of the paper, based on the research strategy you picked?
 - It would be very good if you can **list threats in the four common categories** we usually work with in software engineering.

There is abundant observational data in the software engineering domain, whereas running large-scale controlled experiments is often practically impossible. Thus, most empirical studies can only report statistical correlations—instead of potentially more insightful and robust causal relations.

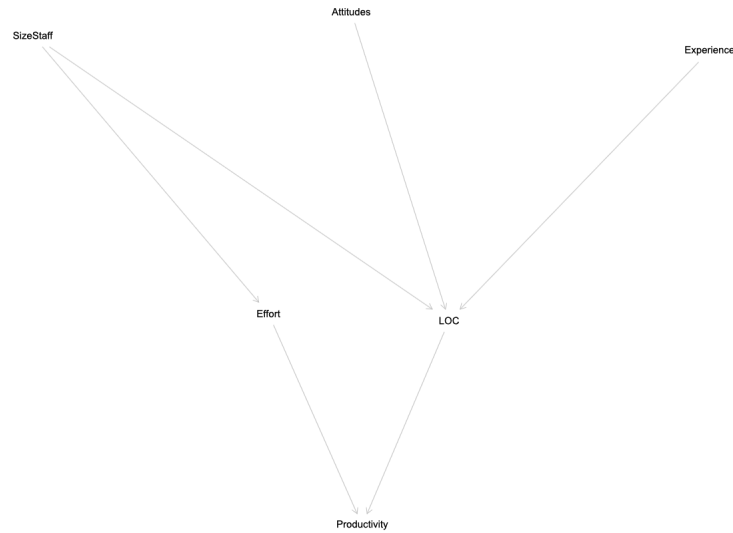
To support analyzing purely observational data for causal relations, and to assess any differences between purely predictive and causal models of the same data, this paper discusses some novel techniques based on structural causal models (such as directed acyclic graphs of causal Bayesian networks). Using these techniques, one can rigorously express, and partially validate, causal hypotheses; and then use the causal information to guide the construction of a statistical model that captures genuine causal relations—such that correlation does imply causation.

We apply these ideas to analyzing public data about programmer performance in Code Jam, a large world-wide coding contest organized by Google every year. Specifically, we look at the impact of different programming languages on a participant's performance in the contest. While the overall effect associated with programming languages is weak compared to other variables—regardless of whether we consider correlational or causal links—we found considerable differences between a purely associational and a causal analysis of the very same data.

The takeaway message is that even an imperfect causal analysis of observational data can help answer the salient research questions more precisely and more robustly than with just purely predictive techniques—where genuine causal effects may be confounded.

Question 8 :

(12p) In software engineering research, which has taken inspiration from the social sciences, we often classify threats to validity in four categories. Which are the four categories (4p)? Please **explain** what question(s) each category tries to answer and provide an **example** for each category (8p).

**Question 9 :**

(6p) See the DAG above. We want to estimate the **total** causal effect of *SizeStaff* on *Productivity*. What should we **condition on**? (1p)

Design a **complete** model, in math notation, where the outcome *Productivity* is a count $0, \dots, \infty$, and **include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions! (5p)